Hoja de Trabajo 4

Ejercicio 1.

$y^2 = 4x + 2y$

$y^2 - 2y + 1 = 4x + 1$ $\qquad 2(1/2)^2 = 1$

$(y-1)^2 = 4(x + 1/4) =$ Parábola

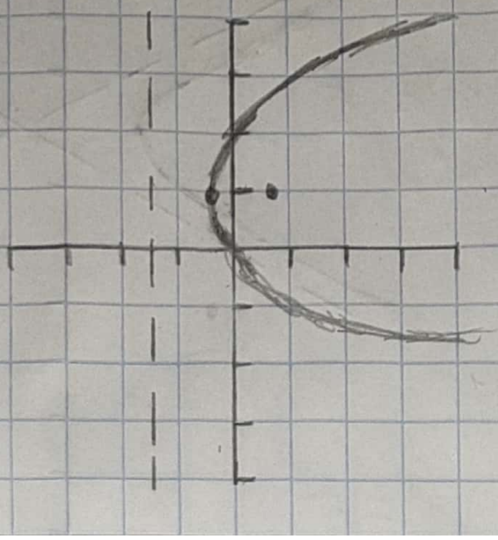
$V = (-1/4, 1)$ $\qquad 4P = 4$

$P = 1$

$F = \left(\frac{3}{4}, 1\right)$

Directriz $= y = k - P$

$D = y = -5/4$



Ejercicio 2.

$x^2 - 4y^2 - 2x + 16y = 20$

$(x^2 - 2x + 1) - 4y^2 + 16y = 20 + 1$

$(x-1)^2 - 4(y^2 + 4y + 4) = 21 - 4*4$

$(x-1)^2 - 4(y-2)^2 = 5$

$\frac{(x-1)^2}{5} - \frac{(y-2)^2}{5/4} = 1$ = Hipérbola

$a = \sqrt{5}$

$b = \frac{\sqrt{5}}{2}$

$c = \sqrt{5 + 5/4}$

$c = 5/2$

Centro = $(1, 2)$

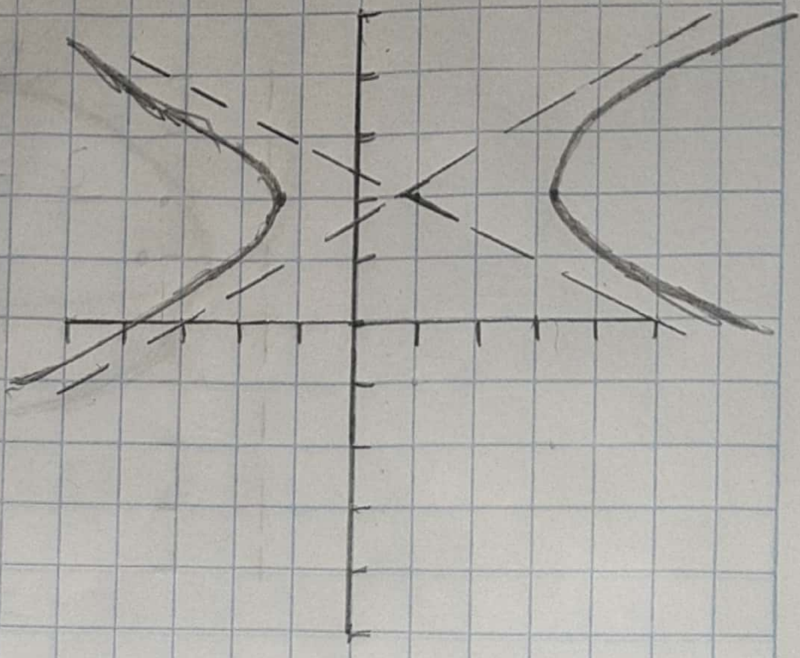
$V_1 = (1 + \sqrt{5}, 2)$

$V_2 = (1 - \sqrt{5}, 2)$

$F_1 = \left(\frac{7}{2}, 2\right)$

$F_2 = \left(-\frac{3}{2}, 2\right)$

Asint. = $y = \pm \frac{1}{2}(x-1) + 2$



Ejercicio 3.

$4X^2+25y^2-24X+250y+561=0$

$4X^2-24X+25y^2+250y=-561$ $\qquad 6(1/2)^2=9$

$4(x^2-6x+9)+25y^2+250y=-561+4(9)$

$4(X-3)^2+25(y^2+250y)=-525$

$4(X-3)^2+25(y^2+10y+25)=-525+25(25)$ $\qquad 10(1/2)^2=25$

$4(X-3)^2+25(y+5)^2=100$

$\dfrac{(X-3)^2}{25}+\dfrac{(y+5)^2}{4}=1$

$a=\sqrt{25}=5$
$b=\sqrt{4}=2$
$c=\sqrt{5^2-2^2}$
$c=\sqrt{21}$

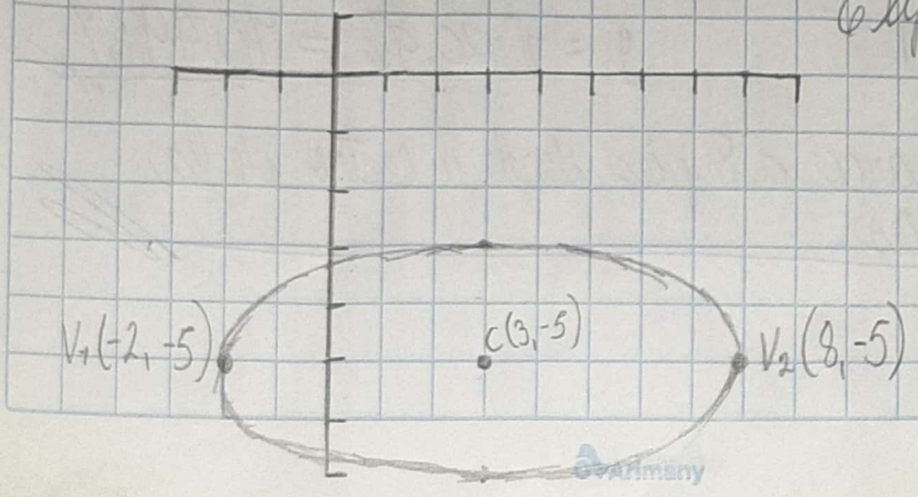
Centro $=(3,-5)$

$V_1=(h-a,K)=(-2,-5)$
$V_2=(h+a,K)=(8,-5)$

$F_1=(h-c,K)=(3-\sqrt{21},-5)$
$F_2=(h+c,K)=(3+\sqrt{21},-5)$

Eje mayor $=5\times2=10$
Eje menor $=2\times2=4$

Elipse



Ejercicio 4.

$x^2 + 16 = 4y^2 + 8x$

$x^2 - 8x + 16 = 4y^2$

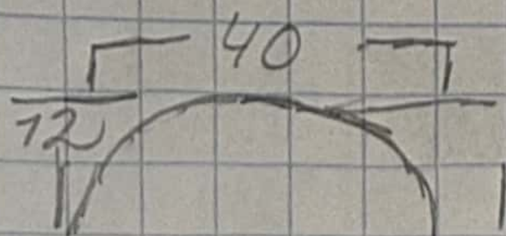
$(x-4)^2 = 4y^2$

Es una cónica degenerada.

Ejercicio 5.

$$\frac{x^2}{20^2}+\frac{y^2}{12^2}=1$$

$$x = \frac{64}{400}+\frac{y^2}{144}=1$$



$$y^2 = 144\left(1-\frac{64}{400}\right)$$

$$y^2 = 120.96$$

$$y = \sqrt{120.96} = \underline{10.9981}$$

La altura del arco a 8 pies desde el centro de la base es 11 pies.